Learning Which Keypoints to Keep: Pair-Label RL Makes ORB and BRISK Match Better, Not Spread Wider

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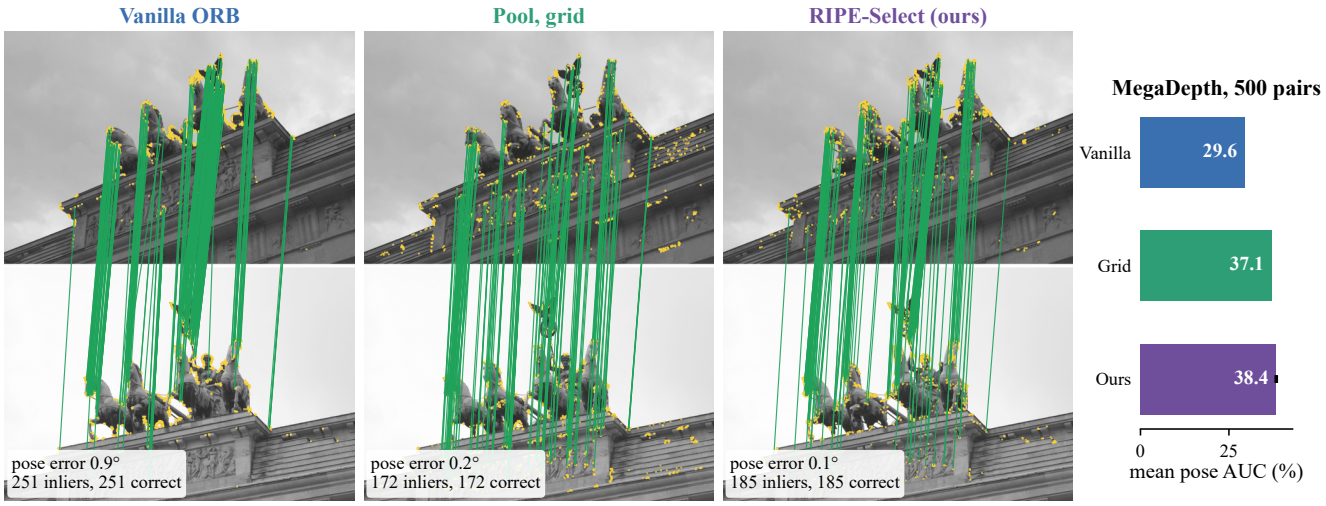


Figure 1. **Same detector, same descriptor, different keypoints.** ORB on a MegaDepth pair, keeping K=2048 keypoints (yellow) with OpenCV's default ranking, a spatial grid, or our learned policy. Lines are RANSAC inliers, green if geometrically correct and orange otherwise. The pair is not hand-picked: its per-pair gain of RIPE-SELECT over vanilla (8.6 AUC points) is the closest to the mean gain over all 500 test pairs (8.7). Right: mean pose AUC over the benchmark.

Abstract

Classical detectors keep the K keypoints with the highest response, but response barely predicts which keypoints will match. We ask whether RIPE's weak reward—a keypoint scores when its match survives RANSAC on a same-scene pair and is penalised on a different-scene pair—can teach a policy which *classical candidates to keep, leaving detector, descriptor and matcher untouched.* RIPE-SELECT *scores candidates with a small MLP and is trained with REINFORCE under a budget-exact Bernoulli selection. Trained on two scenes on a laptop CPU, it improves MegaDepth-1500 pose AUC over OpenCV by* ***+8.8*** *for ORB and* ***+4.7*** *for BRISK (three seeds) and raises match precision; for ORB, RL beats supervised training on the same pseudo-labels by* ***+5.7****. The caveat is spatial: a grid-bucketing heuristic on the same candidates reaches comparable pose accuracy by spreading keypoints, and on planar HPatches no learned variant beats it. Learned selection improves* which *keypoints are kept, not* where.

1. Introduction

Learned local features [5, 8, 18, 20] lead image-matching benchmarks, yet classical detectors remain what many deployed systems run. ORB [16] is the front end of ORB-SLAM [14], and binary features are preferred wherever compute, memory or licensing are tight. These pipelines detect many candidates and keep the K with the highest detector response.

The gap. Response measures local contrast, not *matchability*. On MegaDepth, response separates candidates that become correct matches from those that do not with a ROC-AUC of only 0.56–0.60, and a top-K cut by response discards more than half of the correct matches present in the candidate pool (Fig. 5). The decision of which keypoints to keep is therefore a cheap, unexploited lever.

The idea. RIPE [8] trains a keypoint detector by reinforcement learning from nothing more than whether two images

show the same scene. A classical detector has no heatmap to sample from, so we learn a *selection policy* over its candidates instead (Fig. 2). Detector, descriptor, matcher and robust estimator are unchanged, which makes RIPE-SELECT a drop-in module for existing pipelines.

Contributions.

- RIPE-SELECT, a keypoint selection policy for classical detectors trained with RIPE's pair-label reward, and a budget-exact Bernoulli selection that replaces RIPE's keypoint-count regulariser (Sec. 3).
- A controlled evaluation in which every method shares images, candidate pool, budget, descriptor, matcher and RANSAC, with three seeds and paired bootstrap tests (Sec. 4).
- Gains of +8.8 (ORB) and +4.7 (BRISK) pose AUC over OpenCV on MegaDepth-1500, with higher match precision and inlier ratio (Sec. 5).
- An analysis of what the policy learns and what it misses: it raises per-keypoint quality, beats supervised training for ORB, but does not improve spatial coverage, which a grid heuristic supplies for free (Sec. 6).

Takeaway: *learn quality, but enforce coverage*—the two are complementary, and pose accuracy needs both.

2. Related work

Classical features. SIFT [12] detects difference-of-Gaussian extrema; ORB [16] combines oriented FAST [15] corners, ranked by a Harris score, with rotated BRIEF; BRISK [10] uses AGAST [13] corners and a 512-bit sampling-pattern descriptor. All three control the keypoint count with a response threshold and a top-K cut.

Selection and spatial distribution. Adaptive non-maximal suppression [1, 4] keeps strong keypoints that are far apart; ORB-SLAM [14] buckets keypoints into a grid. These rules change *where* keypoints are kept, not *how good* each is. We use grid bucketing as a baseline and find it a strong one.

Predicting matchability. Hartmann *et al*. [6] classify which SIFT keypoints will match, supervised by structure-from-motion tracks. Our policy needs only image-pair labels and is trained on the outcome of matching and RANSAC itself.

Learned detectors. SuperPoint [5], Key.Net [3], DISK [18] and ALIKED [20] learn detectors from depth, pose or homography supervision; DISK and RIPE [8] use policy gradients, RIPE with only pair labels. These replace the whole feature; we keep the classical one and learn only the selection.

3. Method

Candidate pool. We run the unchanged OpenCV detector with lowered thresholds—FAST threshold 7 with up to 8K points for ORB, AGAST threshold 10 instead of 30 for BRISK—and keep the $4K$ strongest candidates by response. The unchanged OpenCV descriptor is computed for every candidate.

Cues. Each candidate i is described by $\mathbf{f}_i$: 20 scalar cues followed by its descriptor bits (276 inputs for ORB, 532 for BRISK). The cues cover response and its rank, scale and pyramid level, near-duplicate detections, local density and rank in the neighbourhood, intra-image descriptor distinctiveness, and patch statistics (full list in Sec. D). Image coordinates are deliberately excluded, so the policy cannot memorise scene-specific locations.

Budget-exact selection. An MLP ($\to 128 \to 64 \to 1$, ReLU) maps $\mathbf{f}_i$ to a score s_i. During training candidate i is kept, $a_i{=}1$, with probability

$$p_i = \sigma(s_i + b), \qquad b \text{ such that } \textstyle\sum_i p_i = K. \tag{1}$$

Since $\sum_i p_i$ is monotone in b, b is found per image by bisection. The expected number of kept keypoints is thus fixed to the budget: the policy can change *which* keypoints are kept but not *how many*, removing RIPE's keypoint-count regulariser and the quantity–quality trade-off it controls. At test time we keep the top-K scores.

Reward. For a training pair with label $y \in \{\text{same}, \text{diff}\}$, kept keypoints are matched by mutual nearest neighbour (MNN) and a fundamental matrix is fitted with RANSAC [9] at 1.5 px on 1200 px images. Following RIPE, a kept keypoint receives

$$r_i = \begin{cases} +1 & \text{inlier, } y = \text{same} \\ -1 & \text{inlier, } y = \text{diff} \\ 0 & \text{otherwise,} \end{cases} \tag{2}$$

so negative pairs penalise keypoints that produce confident but meaningless matches.

Training. We maximise the expected reward with REINFORCE [19],

$$\nabla_\theta J = \mathbb{E}\Big[\textstyle\sum_i (r_i - \bar{r})\, \nabla_\theta \log \pi_\theta(a_i \mid \mathbf{f}_i)\Big], \tag{3}$$

with the per-image mean reward $\bar{r}$ as baseline. One step is summarised in Algorithm 1.

4. Experimental setup

Training. 350 images each from the IMC-2020 [7] scenes *Palace of Westminster* and *Grand Place Brussels*; positive pairs are covisible pairs from the COLMAP models, negative pairs are drawn across scenes. 1200 steps of 12 pairs, Adam (10^{-3}), 40–90 minutes per run on a laptop CPU. Checkpoints are selected on the held-out scene *Reichstag*; no test image is used for training or selection (hyper-parameters in Sec. E).

Benchmarks. **MegaDepth-1500** [11, 17]: relative pose on scenes 0015 and 0022, AUC of the pose error at 5/10/20°,

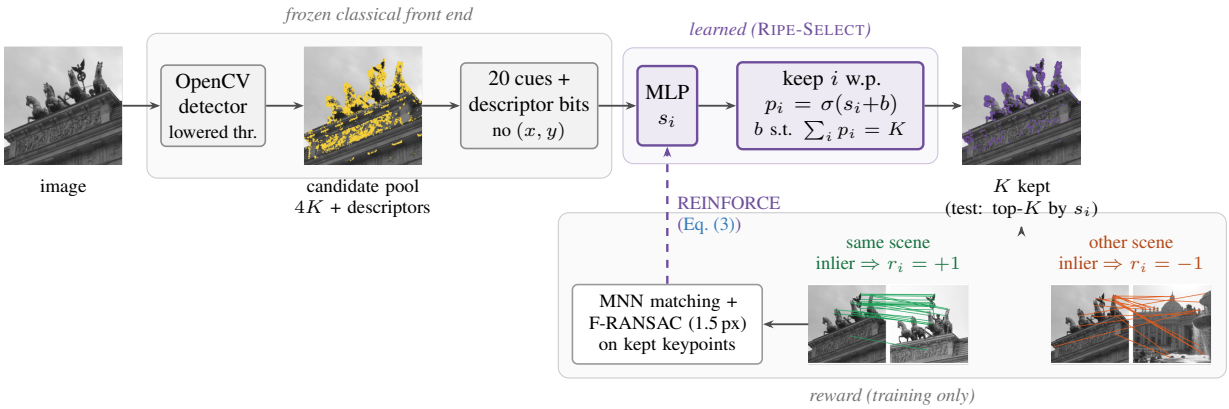


Figure 2. **RIPE-SELECT.** A classical detector run with lowered thresholds proposes a candidate pool; an MLP scores every candidate from 20 local cues and its descriptor, without image coordinates. During training each candidate is kept by a Bernoulli draw whose per-image bias b fixes the expected count to the budget K; kept keypoints are matched and earn $+1$ when they survive RANSAC on a same-scene pair and -1 on a different-scene pair. At test time the top-K scores are kept and passed to the unchanged pipeline.

Algorithm 1: one training step of RIPE-SELECT

Input: 12 image pairs (50 % different-scene), policy π_θ, budget K

1: **for** each image I in the batch **do**
2: pool $\leftarrow$ top-$4K$ detections; $s \leftarrow \pi_\theta(\mathbf{f})$
3: solve b with $\sum_i \sigma(s_i+b) = K$ Eq. (1)
4: sample $a_i \sim \text{Bernoulli}(\sigma(s_i+b))$
5: **for** each pair **do** MNN + F-RANSAC on kept keypoints
6: assign r_i to kept keypoints by pair label Eq. (2)
7: $\bar{r} \leftarrow$ per-image mean of r; update θ with Adam Eq. (3)

K=2048, 500 evenly spaced pairs covering both scenes. **HPatches** [2]: homography on 540 pairs, AUC of the corner error at 1/3/5 px, K=1024.

Protocol. Within a table every row shares the image, candidate pool, budget, OpenCV descriptor, MNN matcher and PoseLib RANSAC [9] (100k iterations, fixed seed). The RANSAC threshold is tuned for vanilla OpenCV and reused for all rows, which favours the baseline.

Baselines. *Vanilla*: OpenCV defaults. *Pool, top-K response*: our candidate pool ranked by response. *Pool, grid*: the same pool in a 16×16 grid, taken round-robin by response as in ORB-SLAM. *Supervised*: our network trained with binary cross-entropy on the RANSAC inlier labels of the full pool, which isolates the effect of RL.

Statistics. RIPE-SELECT rows are means over three seeds (± sample std). Significance is a paired bootstrap over test pairs (2000 resamples, 95 % CI) of the difference in mean AUC.

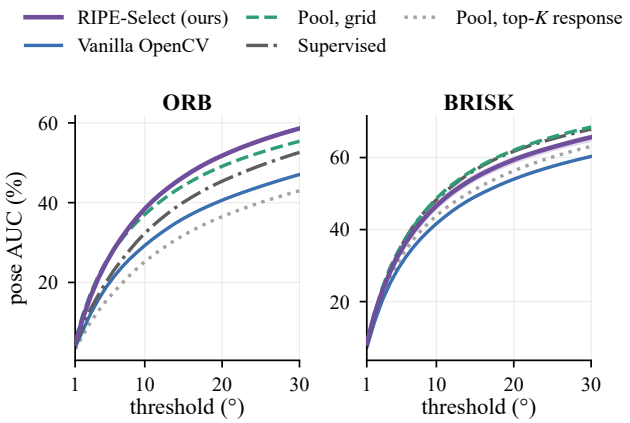


Figure 3. **Pose AUC vs. error threshold** on MegaDepth. Band: range over three RIPE-SELECT seeds.

5. Results

MegaDepth. Table 1 (top) shows that learned selection improves on OpenCV for both detectors, by **+8.8** mean AUC for ORB and **+4.7** for BRISK. The three ORB seeds agree to within 0.2 AUC, the BRISK seeds to within 2.4 (Tab. 4). The improvement holds at every threshold (Fig. 3) and is significant against vanilla and against response ranking of the same pool (Fig. 4). Per pair, RIPE-SELECT improves 47 % of the ORB pairs and worsens 27 % (BRISK: 47 % and 35 %); the rest are unchanged. The spatial grid on the same pool is the strongest baseline: RIPE-SELECT is ahead by 1.2 for ORB and behind by 2.0 for BRISK, neither difference significant.

HPatches. On HPatches (Tab. 1, bottom) RIPE-SELECT is on par with vanilla (-0.1 for ORB, $+0.3$ for BRISK) and significantly below the grid (Fig. 4). The scenes are planar

	ORB				BRISK			
MegaDepth-1500, relative pose (K=2048)								
	@5°	@10°	@20°	Mean	@5°	@10°	@20°	Mean
Vanilla OpenCV	18.8	29.3	40.6	29.6	29.0	41.6	54.0	41.5
Pool, top-K response	15.4	25.2	36.5	25.7	30.8	43.9	56.3	43.7
Pool, grid	**25.3**	37.0	49.1	37.1	**33.8**	**48.7**	**62.1**	**48.2**
Supervised	20.1	32.3	45.4	32.6	33.4	48.3	61.7	47.8
RIPE-SELECT (3 seeds)	24.9	**38.4**	**51.8**	**38.4 ± 0.1**	32.6	46.6	59.4	46.2 ± 1.2
HPatches, homography (K=1024)								
	@1 px	@3 px	@5 px	Mean	@1 px	@3 px	@5 px	Mean
Vanilla OpenCV	21.0	45.0	56.9	41.0	27.5	58.0	69.4	51.6
Pool, top-K response	18.4	42.9	55.1	38.8	28.2	58.6	70.1	52.3
Pool, grid	**24.5**	**51.0**	**62.3**	**45.9**	**31.9**	**61.9**	**72.6**	**55.5**
Supervised	18.1	42.5	54.4	38.3	27.1	57.2	69.0	51.1
RIPE-SELECT (3 seeds)	20.4	45.2	56.9	40.9 ± 0.4	28.1	58.0	69.6	51.9 ± 0.1

Table 1. **Main results.** AUC (%) of the pose error on MegaDepth-1500 (500 pairs) and of the homography corner error on HPatches. Every row uses the same images, budget, descriptor, matcher and RANSAC; the pool rows, supervised and RIPE-SELECT select from the same candidate pool. RIPE-SELECT: mean over three seeds, ± std of the mean AUC. **Bold**: best, underlined: second best.

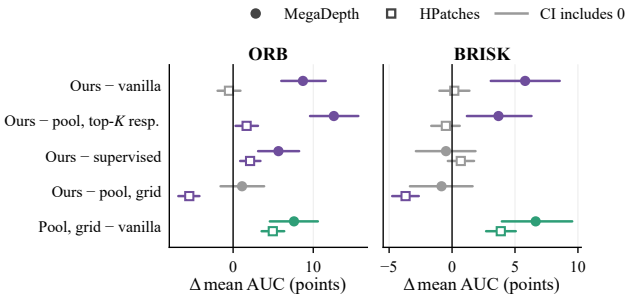


Figure 4. **Paired bootstrap** differences in mean AUC with 95 % CIs (seed-0 policy). Grey: the interval includes zero.

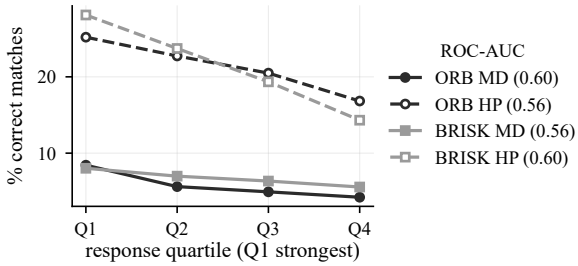


Figure 5. **Response is a weak cue.** Share of candidates that become correct matches, by response quartile (MD: MegaDepth, HP: HPatches); legend: ROC-AUC of response as a predictor.

and well textured, so almost every textured keypoint is valid and selection matters little; the policy was also trained at 1200 px on outdoor photographs, while HPatches images are about 480 px.

Published numbers. Our protocol reproduces the published SIFT row of RIPE [8] on MegaDepth to within 1.4 AUC at 5° (38.7 *vs*. 37.3). RIPE itself, which learns detector and descriptor, stays far ahead (55.1 AUC@5°): RIPE-SELECT can only choose among the classical candidates.

6. Analysis

Is there anything to learn? Figure 5 bins pool candidates by response quartile: on MegaDepth the share that becomes a correct match only falls from 8 % to 4–6 % between the strongest and the weakest quartile, and top-K by response keeps only 40–44 % of the pool's correct matches.

Is it the larger pool? No. Ranking the same pool by response *lowers* ORB accuracy by 3.9 AUC, while the learned ranking raises it by 8.8; RIPE-SELECT beats pool top-K by +12.6 (ORB) and +3.7 (BRISK), both significant (Fig. 4).

Does RL matter? For ORB, RIPE-SELECT beats the supervised model trained on the same RANSAC pseudo-labels by +5.7 [+3.1, +8.2]; for BRISK the two are equal (−0.5 [−2.9, +1.9]). An obvious explanation is the negative-pair penalty, which supervised labels never see. We tested it on 50 MegaDepth pairs, each also matched against an image of the other scene, and it is *not* supported: for ORB, RIPE-SELECT lowers the mean number of RANSAC inliers on these negative pairs only from 21.8 to 19.5, and the ratio of inliers on positive to negative pairs is unchanged (7.0 *vs*. 7.1). For BRISK the ratio improves (8.4 *vs*. 6.2) but RL and supervised are equal. Why RL helps ORB remains open; one untested explanation is that RL learns from matches among *sampled* subsets of the pool, as at test time, whereas the supervised labels are computed once on the full pool.

Quality or coverage? RIPE-SELECT and the grid reach similar pose accuracy by opposite routes (Tab. 2). The grid

		MMA (%)			Inl.	Cov.	Time
		@1 px	@3 px	@5 px	(%)	(%)	(ms)
ORB	Vanilla	16.9	27.4	30.0	**24.0**	7.6	**20**
	Pool, grid	12.2	20.1	22.2	17.6	**12.3**	70
	RIPE-SELECT	**17.0**	**27.5**	**30.1**	**24.0**	9.7	838
BRISK	Vanilla	16.7	24.7	27.1	22.2	10.6	**81**
	Pool, grid	15.0	22.3	24.3	20.2	**17.0**	295
	RIPE-SELECT	**18.1**	**26.9**	**29.2**	**24.5**	14.6	1077

Table 2. **Quality, coverage and cost** on MegaDepth (seed 0). MMA: share of MNN matches with epipolar error below the threshold. Inl.: RANSAC inlier ratio. Cov.: share of 16×16 cells containing a RANSAC inlier (50 pairs). Time: median single-thread detection and selection per image, measured in isolation.

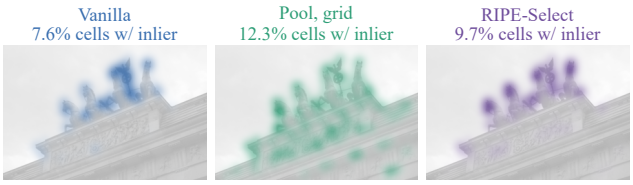


Figure 6. **Where the kept keypoints are.** Density of the 2048 kept ORB keypoints on the first image of Fig. 1. Titles: share of 16×16 cells containing a RANSAC inlier, averaged over 50 MegaDepth pairs.

lowers match precision and inlier ratio (ORB inlier ratio 24.0 % → 17.6 %) but spreads its inliers over the image: 12.3 % of grid cells contain an inlier for ORB, against 7.6 % for vanilla. RIPE-SELECT keeps or raises precision (BRISK inlier ratio 22.2 % → 24.5 %) and covers more than vanilla but less than the grid (9.7 %; Fig. 6). With no coordinates as input and a per-keypoint reward, the policy has no incentive to cover the image, which robust pose estimation also rewards. A coverage-aware reward is the natural next step.

What does it cost? The policy needs the larger pool and the cues. In an isolated single-thread benchmark (median of 20 MegaDepth images) the pipeline takes 838 ms per image for ORB and 1077 ms for BRISK, against 20 and 81 ms for OpenCV—42× and 13× (Fig. 7). About 740 ms of this is cue computation in unoptimised Python; the MLP takes 27–41 ms and the final selection about 1 ms. The grid costs only the pool extraction (70 and 295 ms), which makes it the better accuracy–cost trade-off today.

7. Limitations

Training uses two scenes (700 images), no hyper-parameter search and one validation scene; RIPE uses the DISK MegaDepth subset and about three GPU-days. MegaDepth results use 500 of the 1500 pairs; RL rows use three seeds, and the ablations and bootstrap tests use seed 0. Cue computation makes selection slow. For SIFT and SURF learned selection does not help (Sec. A).

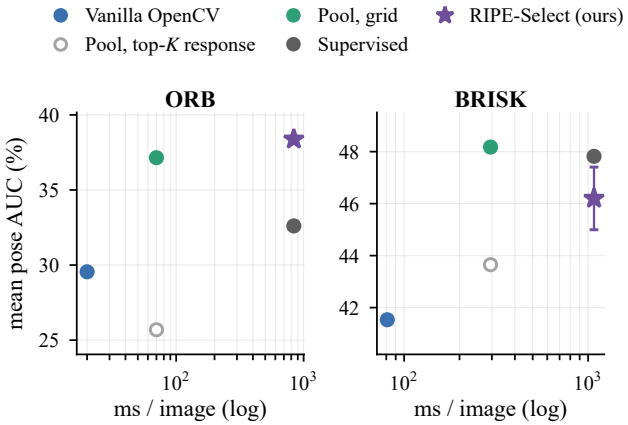


Figure 7. **Accuracy vs. cost** on MegaDepth. RIPE-SELECT: three-seed mean ± std; supervised shares its runtime.

8. Conclusion

RIPE's pair-label reward transfers to classical features as a keypoint selection policy. Without changing detector, descriptor or matcher, RIPE-SELECT raises match precision and improves relative pose on MegaDepth for ORB and BRISK, and for ORB reinforcement learning clearly beats supervised training on the same labels. A free spatial heuristic, however, reaches similar pose accuracy through coverage rather than quality. Learning quality while enforcing coverage—in the reward or in the selection rule—is the clear next step.

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Supplementary Material

Learning Which Keypoints to Keep

A. SIFT and SURF

We ran the same pipeline for SIFT and SURF; neither supports a claim that learned selection helps, which is why the main paper covers ORB and BRISK (Tab. 3). **SIFT.** The three-seed mean improves on vanilla by +3.3 mean AUC on MegaDepth, but the gain depends on the seed (+6.4, +3.3 and +0.1 for seeds 0–2) and is not significant across seeds (p=0.21, one-sample t-test on the seed gains). About 2.4 of the 3.3 points come from ranking the larger pool by response; the grid gains more (+5.1, significant), and the supervised model is at least as good as RL. On HPatches RL selection is significantly worse than vanilla (-1.4 [-2.4, -0.3]). SIFT's response is already more informative (the correct-match rate falls from 10.4 % to 6.0 % over the quartiles), leaving less room for learned selection. **SURF.** Learned selection hurts on MegaDepth: -5.3 mean AUC against vanilla for seed 0, also for supervised training and for a residual policy initialised at the response ranking. On the validation scene the supervised model had helped (+3.6), so the policy overfits to the training landmarks.

MegaDepth AUC @5/10/20°	SIFT	SURF
Vanilla OpenCV	38.7 / 51.8 / 63.2	**33.7 / 47.0 / 60.2**
Pool, top-K response	40.6 / 54.1 / 66.2	33.0 / 46.7 / **60.2**
Pool, grid	**43.3 / 56.9 / 68.7**	31.3 / 44.7 / 57.8
Supervised	42.5 / **57.1 / 69.9**	28.6 / 42.9 / 58.0
RIPE-SELECT (SIFT: 3 seeds; SURF: seed 0)	41.0±2.4 / 55.3±3.3 / 67.3±3.7	27.0 / 41.8 / 56.1

Table 3. SIFT and SURF on MegaDepth-1500 (500 pairs, K=2048). Bold: best per column.

B. Per-seed results

		MegaDepth AUC			HPatches AUC			Val.
		@5°	@10°	@20°	@1 px	@3 px	@5 px	AUC
ORB	seed 0	25.3	38.2	51.3	20.3	44.8	56.3	35.1
	seed 1	25.0	38.6	52.0	20.3	45.2	57.3	34.3
	seed 2	24.5	38.5	52.0	20.7	45.6	57.3	34.5
BRISK	seed 0	33.9	47.9	60.2	28.3	57.9	69.2	59.5
	seed 1	31.3	45.3	58.2	27.8	58.2	70.2	60.0
	seed 2	32.6	46.7	59.7	28.3	57.9	69.5	58.3

Table 4. RIPE-SELECT per training seed. Val.: best mean AUC on the held-out *Reichstag* scene, used for checkpoint selection (plain response: ORB 17.2, BRISK 46.5).

C. Additional qualitative results

Figure 8 shows the BRISK pair whose per-pair gain is closest to the BRISK mean gain; Fig. 9 shows an ORB pair on which RIPE-SELECT is worse than vanilla, chosen as the pair whose loss is closest to the mean ORB gain with opposite sign. In the failure case all three selections recover the pose roughly (8.5–10.6°), and RIPE-SELECT has more correct inliers but concentrated in a smaller region.

D. Per-candidate cues

The 20 scalar cues computed for every pool candidate: (1) log response; (2) log response standardised over the image; (3) response rank in the pool; (4) log size; (5) pyramid octave; (6) layer within the octave (SIFT only, otherwise 0);

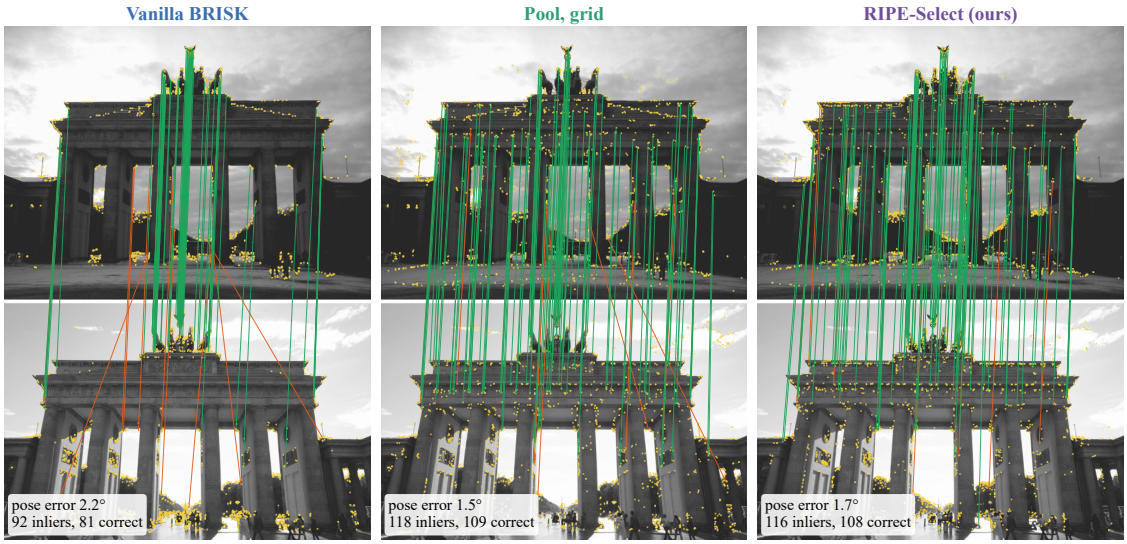


Figure 8. BRISK, representative pair (per-pair gain 5.8 *vs*. mean 5.8 AUC points).

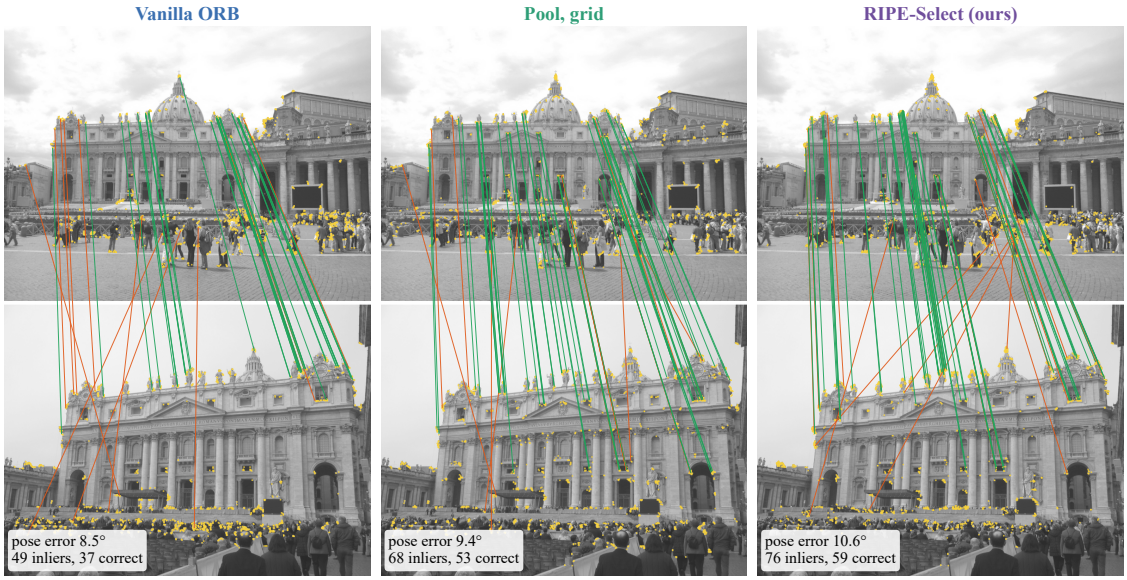


Figure 9. ORB, failure case (per-pair change -8.6 AUC points).

(7) number of other detections within 0.5 px; (8) number of candidates within 16 px; (9) response rank among those neighbours; (10, 11) distance to the nearest and second-nearest other descriptor in the same image, normalised by the image median; (12) their ratio; (13–15) mean intensity, intensity standard deviation and mean gradient magnitude in a box of radius $0.75\times$ size; (16, 17) mean and standard deviation relative to the whole image; (18) log pool size; (19, 20) image mean intensity and standard deviation. They are followed by the descriptor as ± 1 bits (256 for ORB, 512 for BRISK).

E. Hyper-parameters and protocol

Pool	$4K$ candidates (8192 at K=2048); ORB: FAST thr. 7, up to 8K; BRISK: AGAST thr. 10 (default 30)
Policy	MLP, inputs $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 1, ReLU
Training	1200 steps $\times$ 12 pairs, 50 % negative, K=2048, Adam with learning rate 10^{-3}
Reward RANSAC	fundamental matrix, 1.5 px at 1200 px, 2000 iterations
REINFORCE baseline	per-image mean reward
Validation	every 100 steps, 96 pairs of *Reichstag*
Test RANSAC	PoseLib, 100k iterations, seed 0
RANSAC threshold	tuned on vanilla: MegaDepth 1.0 px (from 1.0, 1.5); HPatches ORB 2 px, BRISK 4 px (from 1, 2, 3, 4, 6)
Runtime	single thread, warm-up pass, median over 20 images (`timing_benchmark.py`)

Table 5. Training and evaluation settings.